TC2343

LAB02 GAZEBO SIMULATION

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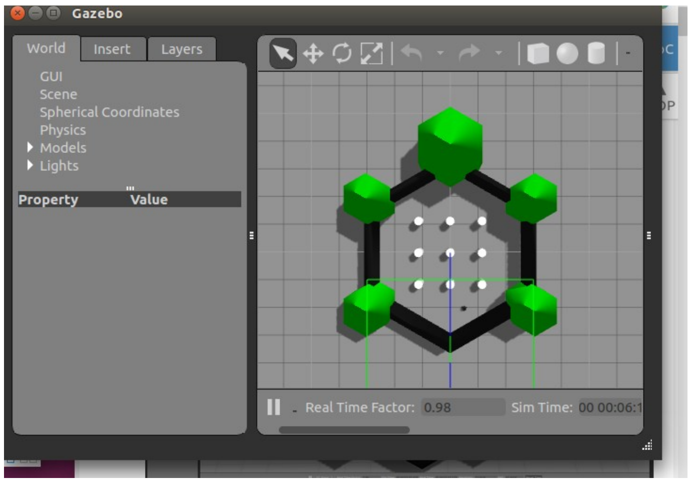
FTSM UKM

Marks: 20 marks (Lab assignment 2)

Instruction

1. Run turtlebot3 basic simulation. Follow the instruction in link below:

    http://emanual.robotis.com/docs/en/platform/turtlebot3/simulation/



a. Launch turtlebot burger and waffle_pi using turtlebot3_world.launch instruction

b. Drive turtlebot 3 using teleop

c. Execute Rviz to visualize published topic, use turtlebot model "waffle_pi" to get Image view (turtlebot3 burger don't have camera).

c. Observe and the following:

i- view ROS graph	**$ rqt_graph**
ii- rostopic command	**$ rostopic**
iii- check active rostopic	**$ rostopic list**
iv- view example of data	**$ rostopic echo /scan**

v- observe and record cmd_vel data value when moving the turtlebot3

- front, back, right, left, stop

2. Run Virtual SLAM and navigation with TurtleBot3 until you complete the whole map and save

3. Run Virtual SLAM by Multiple TurtleBot3s and merge map from all three turtlebots

Tasks

Write a report and discuss the following:

1. Rosgraph for all 3 scenarios above. Display the rosgraph and explain your observations.
2. Explain differences using ROS graph between turtlebot3 burger and waffle pi.
3. View and discuss the content of launch file (*.launch) for all three scenarios above.
4. Try to move the turtlebot3 autonomously in a circle and square using python code. Pleasesubmit before next class. (Referennce: http://wiki.ros.org/turtlesim)

Please submit the assignment in UKMFolio before next lab session.

"HAPPY ROBOTING"

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